

Triangle Inequalities and Theorems

Geometry Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Apply the Comparison Property of Inequalities to triangle side lengths and angles
- Use the Exterior Angle Inequality Theorem to compare angles in a triangle
- Apply the Isosceles Triangle Theorem to prove angle relationships

Problems

1. In a triangle, if $a = b + c$ and $c > 0$, which inequality must be true?

$$a = b + c, c > 0$$

2. Angle 1 is an exterior angle of a triangle with remote interior angles 2 and 3. If $m\angle 2 = 55^\circ$ and $m\angle 3 = 48^\circ$, find $m\angle 1$.

$$m\angle 1 = m\angle 2 + m\angle 3$$

3. An exterior angle of a triangle measures 112° . One remote interior angle measures 67° . Find the measure of the other remote interior angle.

$$112^\circ = 67^\circ + x$$

4. In triangle ABC, exterior angle ACD is formed by extending side BC. Which two angles is $m\angle ACD$ greater than?

$$m\angle ACD > m\angle A \text{ and } m\angle ACD > m\angle B$$

5. In isosceles triangle PQR, $PQ \cong PR$. What can you conclude about the base angles?

$$PQ \cong PR \Rightarrow m\angle Q = m\angle R$$



6. In triangle XYZ, an exterior angle at vertex Z measures 95° . Can the remote interior angles each measure 50° ? Explain.

$$50^\circ + 50^\circ = 100^\circ \neq 95^\circ$$

7. In the figure, point M lies on segment AC of triangle ABC such that $BM \cong BC$. If $m\angle ABM = 40^\circ$, find $m\angle BMC$ (the exterior angle of triangle ABM at M).

$$m\angle BMC = m\angle ABM + m\angle BAM$$

8. In triangle DEF, extend side EF to point G. Using the Exterior Angle Inequality Theorem, write two inequality statements involving the exterior angle DFG.

$$m\angle DFG > m\angle D \text{ and } m\angle DFG > m\angle E$$

9. In triangle ABC, point D is the midpoint of BC such that AD is drawn, creating triangles ABD and ACD. $AB \cong AC$ (isosceles). Using the Isosceles Triangle Theorem and Exterior Angle Theorem, prove that $m\angle ADB > m\angle ABD$.

$$m\angle ADB = m\angle DAC + m\angle ACD$$

10. In triangle PQR, S is a point on QR such that PS is drawn. Given that $PQ \cong PS$, $m\angle QPR = 80^\circ$, and $m\angle PSR$ is an exterior angle of triangle PQS, find $m\angle PSR$.

$$m\angle PSR = m\angle PQS + m\angle QPS$$



Triangle Inequalities and Theorems — Answer Key

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Answer Key

1. Answer: $a > b$

- Since $c > 0$, adding c to b makes the sum strictly greater than b alone.
- Therefore $a = b + c > b$.

2. Answer: 103°

- By the Exterior Angle Theorem, $m\angle 1 = m\angle 2 + m\angle 3$.
- $m\angle 1 = 55^\circ + 48^\circ = 103^\circ$.

3. Answer: 45°

- By the Exterior Angle Theorem, the exterior angle equals the sum of the two remote interior angles.
- $x = 112^\circ - 67^\circ = 45^\circ$.

4. Answer: $m\angle ACD$ is greater than both $m\angle A$ and $m\angle B$

- By the Corollary to the Exterior Angle Theorem, an exterior angle is greater than each of its two remote interior angles.
- Therefore $m\angle ACD > m\angle A$ and $m\angle ACD > m\angle B$.

5. Answer: $m\angle Q = m\angle R$ (base angles are congruent)

- By the Isosceles Triangle Theorem, if two sides of a triangle are congruent, the angles opposite those sides are congruent.
- Therefore $m\angle Q = m\angle R$.

6. Answer: No, because $50^\circ + 50^\circ = 100^\circ$, not 95°

- The Exterior Angle Theorem requires the exterior angle to equal the sum of the two remote interior angles.
- Since $50^\circ + 50^\circ = 100^\circ \neq 95^\circ$, this is impossible.

7. Answer: Cannot be determined without $m\angle BAM$; if $m\angle BAM = 35^\circ$, then $m\angle BMC = 75^\circ$

- By the Exterior Angle Theorem, $m\angle BMC = m\angle ABM + m\angle BAM$.
- Substitute the given values: $m\angle BMC = 40^\circ + m\angle BAM$.

8. Answer: $m\angle DFG > m\angle D$ and $m\angle DFG > m\angle E$

- $\angle DFG$ is an exterior angle of triangle DEF at vertex F.
- By the Exterior Angle Inequality Theorem, it is greater than each remote interior angle: $m\angle D$ and $m\angle E$.

9. Answer: $m\angle ADB > m\angle ABD$ because $m\angle ADB$ is an exterior angle of triangle ACD and exceeds each remote interior angle

- Since $AB \cong AC$, by the Isosceles Triangle Theorem $m\angle ABD = m\angle ACD$.



- $\angle ADB$ is an exterior angle of triangle ACD , so $m\angle ADB = m\angle DAC + m\angle ACD > m\angle ACD = m\angle ABD$.

10. Answer: $m\angle PSR = 90^\circ$

- Since $PQ \cong PS$, triangle PQS is isosceles, so $m\angle PQS = m\angle PSQ$. The angles of triangle PQS sum to 180° , with $m\angle QPS$ being part of $m\angle QPR = 80^\circ$.
 - If we take $m\angle QPS = 80^\circ$ (angle at P in triangle PQS), then $m\angle PQS = m\angle PSQ = (180^\circ - 80^\circ)/2 = 50^\circ$. The exterior angle $m\angle PSR = m\angle PQS + m\angle QPS = 50^\circ + 80^\circ/\dots$ Using exterior angle theorem: $m\angle PSR = 180^\circ - 50^\circ = \dots$ By the exterior angle theorem $m\angle PSR = m\angle PQS + m\angle QPS = 50^\circ + 40^\circ = 90^\circ$.
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